App Development 

Brownie Coding for Good Badge — Girls Build Bots

Presented by your robotics mentors!

We’re not just coding today 💚

We’re **solving a problem for a real person**.
That’s what “coding for good” means.

Apps are problem-solvers 🔧

Think about apps you know. For each one ask:

Whose problem does this solve?

- A weather app → helps you know what to wear ☔
- A maps app → helps you not get lost
- A timer → helps you remember 🍳

Apps aren't magic. **Someone noticed a problem and built a fix.**

Find a problem worth solving 🔍

What's a problem in your **school**, **troop**, **home**, or **neighborhood**?

- Remembering to water the class plants 🌱
- Finding lost library books 📚
- Knowing when the bus comes 🚌

No idea is too small. Pick ONE problem your app will solve.

Design your app on paper ✏

This is how **real app designers** start!

Screen 1 — Welcome: your app's name + what it does
Screen 2 — Main screen: the buttons that solve the problem **Screen 3** — What happens when you tap the main button

Paper apps don't need to work. They need to show the **idea**.

Mentor question

As you design, we'll keep asking you one thing:

"What happens when someone taps here?"

The Click Test 👆

With a partner:

1. One of you **uses** the paper app — point at the buttons
2. The designer **swaps the screens** by hand
3. Does it make sense? Fix it!

Real designers call this a **click-through test**.

Robot moment 

Before our team builds the robot's control app, we decide **what it needs to do** and **who's going to use it**.

Same thinking as yours: *who needs this, and what does it do?*

Share your app ✨

In one sentence:

“My app is called ____ and it helps ____.”

You did it! 🎉

Today you: - ✅ Learned that apps solve problems for people - ✅ Found a real problem in your community - ✅ Designed a 3-screen paper prototype - ✅ Showed how it works with a click test

You are an app designer now!

Keep going at home! 

- Keep a **problem list** — what bugs you? What could an app fix?
- Design more paper screens for your app
- Ask your troop leader about the next Coding for Good badge!

📣 Share Your Impact

Troop leaders & robotics teams — if you used this curriculum, please:

1. **Credit us** — the curriculum comes from **Girls Build Bots**
2. **Tag us** — **@girlsbuildbots** on Instagram
3. **Report your results** — email **girlsbuildbots@gmail.com** with how many girls took part, how many badges were earned, and how it went

This is how the program grows — and how the next troop and team get better materials.